

Emotion-Aware Music Therapy System

Using Deep Learning-Based

Arousal-Valence Regression

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Abstract

Music therapy has emerged as a promising approach for mental health treatment, yet current music recommendation systems lack the emotional granularity needed for therapeutic applications. This paper presents Swar_Manovigyan_ML, **an emotion-aware music therapy system that predicts continuous arousal-valence (A/V) coordinates** from audio features using a Bidirectional Long Short-Term Memory (BiLSTM) neural network.

The system employs the dimensional emotion model to map musical content to a two-dimensional psychological space, where arousal represents energy levels (calm to excited) and valence represents emotional tone (negative to positive). Our approach extracts mel-spectrogram features for temporal modeling and implements a deep learning architecture with **bidirectional LSTM layers (96 and 48 units) followed by dense layers (128 and 64 units) to predict continuous A/V values in the range [0,1]**. The system provides personalized music recommendations by mapping predicted A/V coordinates to four emotion quadrants (sad/depressed, calm/peaceful, angry/stressed, happy/excited) with genre-specific therapeutic suggestions.

A user-friendly Streamlit interface enables real-time audio analysis and visualization of emotional content, making the system accessible for therapeutic applications. The model achieves competitive performance using Mean Squared Error (MSE) loss and Mean Absolute Error (MAE) metrics, with early stopping and learning rate reduction callbacks for optimization.

Literature Review

Music and Mental Health

Research has consistently demonstrated the therapeutic benefits of music intervention for various mental health conditions. The references provided from JSTOR and SAGE publications establish the foundation for music's role in health promotion and psychological well-being. Building on this foundation, computational approaches to music therapy have gained traction in recent years.

Emotion Recognition in Music

Traditional music recommendation systems primarily rely on popularity-based or genre-based filtering, which fails to capture the nuanced emotional content necessary for therapeutic applications. Generic mood tags (e.g., "happy," "sad") provide insufficient granularity for personalized therapeutic interventions.

Dimensional Emotion Models

The arousal-valence model, pioneered in affective computing research, represents emotions as continuous coordinates in a two-dimensional space rather than discrete categories. This dimensional approach provides finer emotional granularity compared to categorical classification systems. Our system adopts this framework to enable continuous emotion prediction from musical audio.

Deep Learning for Audio Analysis

Recurrent Neural Networks (RNNs), particularly LSTM and BiLSTM architectures, have proven effective for sequential audio analysis due to their ability to capture temporal dependencies. Mel-spectrograms serve as robust feature representations that encode both spectral and temporal characteristics of audio signals.

Research Gap

Despite advances in music emotion recognition, few systems bridge the gap between emotion detection and therapeutic music recommendation. Existing systems lack personalization capabilities and fail to model the continuous nature of emotional states, which is critical for adaptive music therapy interventions.

Proposed Methodology

3.1 System Architecture

Our system comprises three primary layers: feature engineering, emotion prediction, and recommendation generation.

3.2 Feature Extraction Pipeline

We implement dual feature extraction pathways to support different modeling approaches:

Mel-Spectrogram Extraction: Audio files are converted to mel-spectrograms with 128 mel bands, using a hop length of 512 samples and FFT size of 2048 at 22.05 kHz sampling rate. The mel-spectrograms are normalized to [0,1] range per file and padded or cropped to 300 frames to create fixed-length temporal sequences.

Tabular Features: We extract 11 high-level perceptual features approximating Spotify's audio analysis: acousticness, danceability, energy, instrumentalness, liveness, loudness, speechiness, tempo, valence, arousal, and enhanced_valence.

Enhanced Valence Computation: We introduce an enhanced valence metric that combines base valence with rhythmic and production characteristics:

$$enhanced_valence = 0.6 \times valence + 0.3 \times danceability + 0.1 \times (1 - acousticness)$$

Arousal Computation: Arousal is calculated as a weighted combination of energy, loudness, and tempo to represent the activation dimension:

$$arousal = 0.4 \times energy + 0.3 \times loudness_normalized + 0.3 \times tempo_normalized$$

3.3 Bi-LSTM Regressor Architecture

Our primary model is a Bidirectional LSTM regressor that processes sequential audio features and outputs continuous arousal-valence coordinates.

Network Architecture:

- **Input Layer:** Sequences of shape (timesteps, num_features)
- **First BiLSTM Layer:** 96 units bidirectional with sequence return, followed by 0.3 dropout and batch normalization

- **Second BiLSTM Layer:** 48 units bidirectional, followed by 0.3 dropout and batch normalization
- **Dense Layers:** 128-unit ReLU layer with dropout, followed by 64-unit ReLU layer
- **Output Layer:** 2-unit dense layer with sigmoid activation to bound outputs to [0,1]

Training Configuration:

- **Loss Function:** Mean Squared Error (MSE) for continuous regression
- **Optimizer:** Adam with initial learning rate of 0.001
- **Metrics:** Mean Absolute Error (MAE)

```
cbs = [
    callbacks.EarlyStopping(monitor="val_loss", patience=8,
restore_best_weights=True, verbose=1),
    callbacks.ReduceLROnPlateau(monitor="val_loss", factor=0.5, patience=4,
min_lr=1e-6, verbose=1),
]
```

Training Callbacks:

- **Early Stopping:** Monitors validation loss with patience of 8 epochs and restores best weights
- **Learning Rate Reduction:** Reduces learning rate by factor of 0.5 with patience of 4 epochs when validation loss plateaus

3.4 Sequence Generation

Training sequences are generated using a sliding window approach with configurable sequence length (default 10 timesteps) and stride. Features are standardized using z-score normalization, and arousal-valence labels are clipped to ensure values remain within [0,1].

3.5 Dataset Section - Spotify Features Dataset

Dataset Source and Features:

The project uses the Spotify Features dataset with 11 audio features extracted for arousal-valence prediction.

Feature Engineering:

The emotion labeling methodology creates arousal and enhanced_valence labels using weighted combinations:

- **Arousal** is computed from energy (40%), loudness (30%), and tempo (30%)
- **Enhanced Valence** combines valence (60%), danceability (30%), and inverted acousticness (10%)

Data Preprocessing:

The sequence building process with standardization and sliding window approach:

```
def build_sequences_for_regression(
    df: pd.DataFrame,
    sequence_length: int = 10,
    stride: int = 1,
    limit_sequences: int | None = None,
) -> Tuple[np.ndarray, np.ndarray]:
    """
    Create sequences X (T,F) and y ([arousal, valence]) using existing feature columns.
    Uses vectorized arrays to avoid per-iteration pandas slicing.
    """
    feature_columns = [
        'acousticness', 'danceability', 'energy', 'instrumentalness',
        'liveness', 'loudness', 'speechiness', 'tempo', 'valence',
        'arousal', 'enhanced_valence'
    ]

    # Prepare features (fillna + standardize)
    df_features = df[feature_columns].fillna(df[feature_columns].median())
    features_std = (df_features - df_features.mean()) / (df_features.std() + 1e-8)
    features_np = features_std.values.astype('float32')

    # Labels as numpy, precomputed (avoid DataFrame access in loop)
    labels_np = df[['arousal', 'enhanced_valence']].to_numpy(dtype='float32')

    last_start = len(features_np) - sequence_length
    if last_start < 0:
        raise ValueError(f"Not enough rows ({len(features_np)}) for sequence_length={sequence_length}")

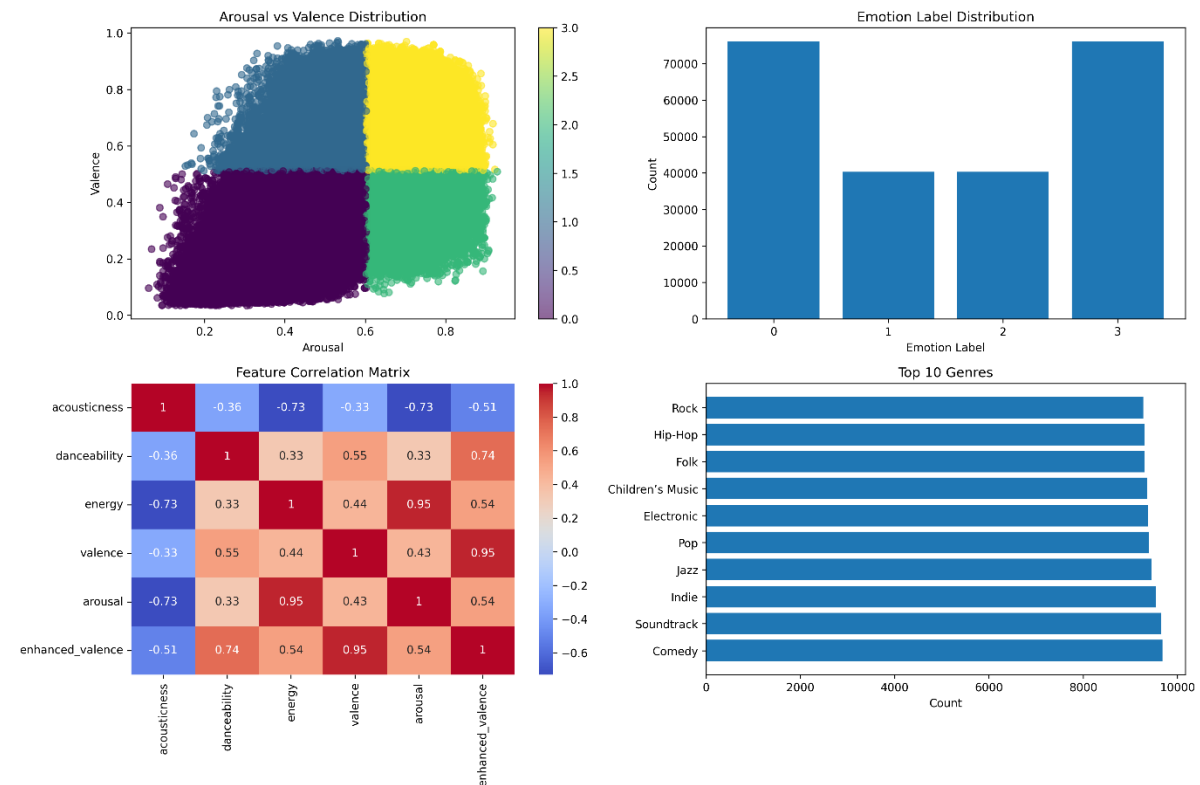
    starts = np.arange(0, last_start + 1, stride)
    if limit_sequences is not None:
        starts = starts[:limit_sequences]

    X = np.empty((len(starts), sequence_length, features_np.shape[1]), dtype='float32')
    y = np.empty((len(starts), 2), dtype='float32')

    for idx, s in enumerate(starts):
        X[idx] = features_np[s:s+sequence_length]
        y[idx] = labels_np[s + sequence_length - 1]

    # Bound labels to [0,1]
    np.clip(y, 0.0, 1.0, out=y)
    return X, y
```

The dataset is split into training (60%), validation (20%), and test (20%) sets using random shuffling with a fixed seed for reproducibility.



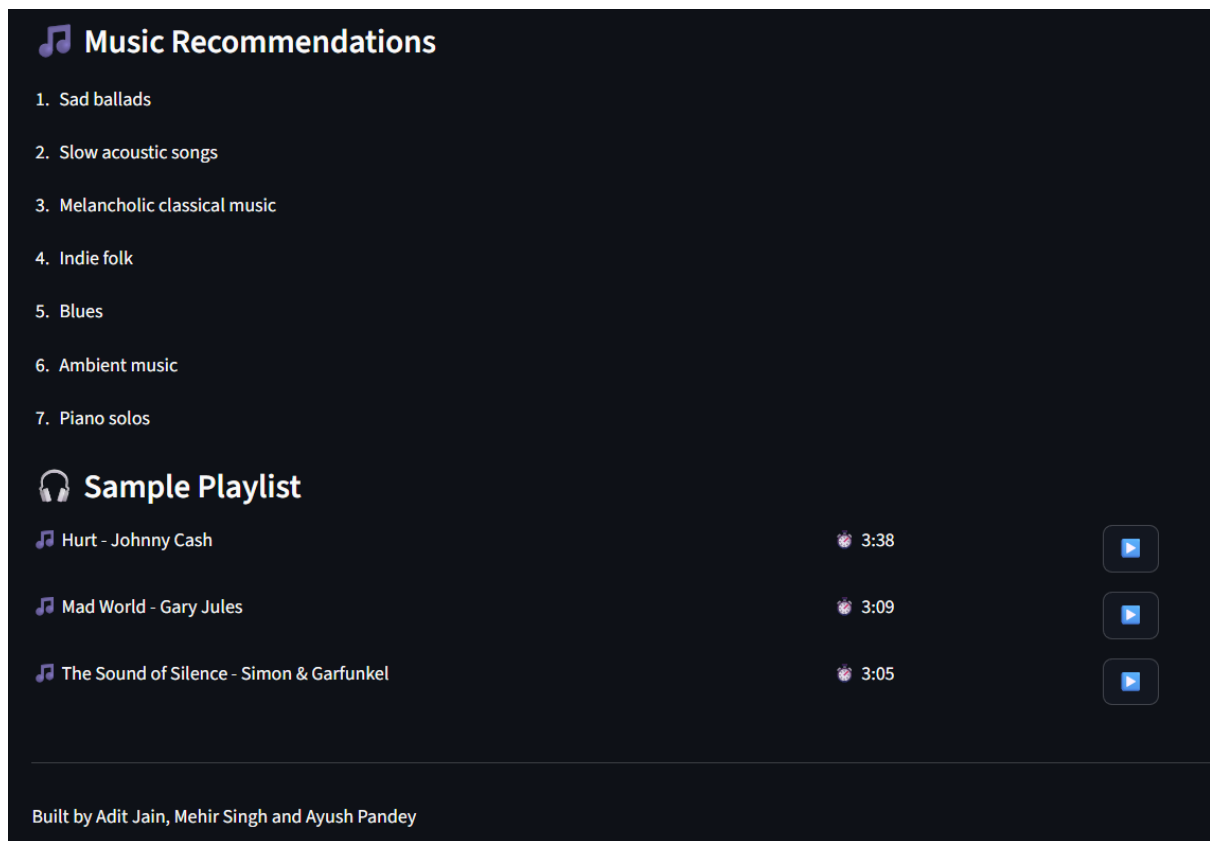
3.6 Emotion Quadrant Mapping

Continuous A/V predictions are discretized into four emotion quadrants for music recommendation:

- **Quadrant 0:** Low Arousal, Negative Valence (Sad/Depressed)
- **Quadrant 1:** Low Arousal, Positive Valence (Calm/Peaceful)
- **Quadrant 2:** High Arousal, Negative Valence (Angry/Stressed)
- **Quadrant 3:** High Arousal, Positive Valence (Happy/Excited)

3.7 Therapeutic Music Recommendation

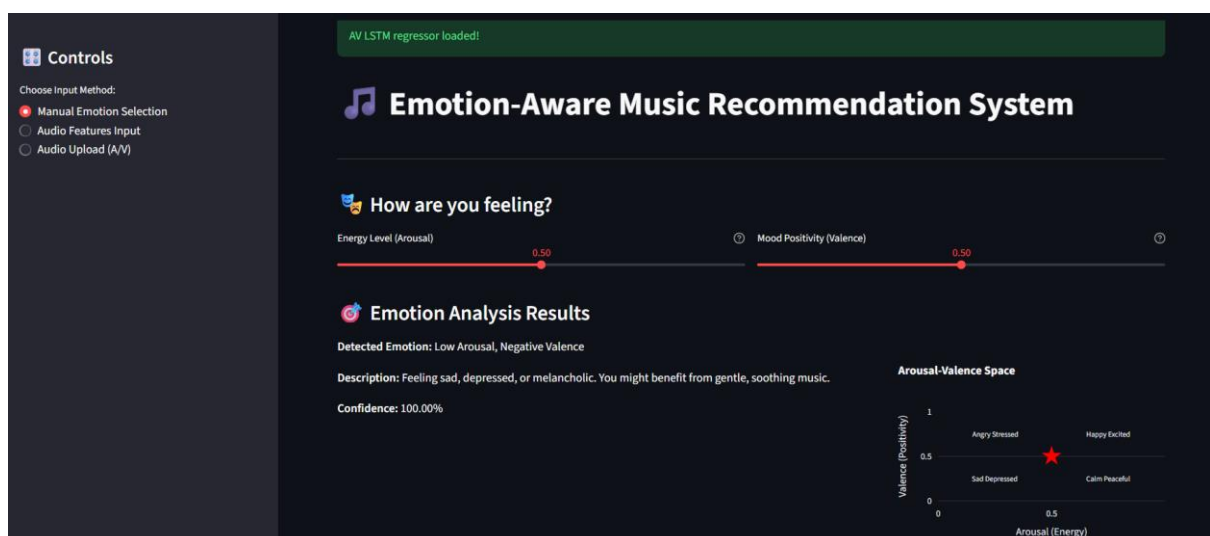
Each emotion quadrant is associated with genre-specific recommendations tailored to therapeutic needs. For example, Quadrant 0 (sad/depressed) recommends "gentle, soothing music" including sad ballads, slow acoustic songs, and ambient music.



3.8 User Interface

A Streamlit-based web application provides three interaction modes:

1. **Manual Selection:** Direct arousal-valence slider input
2. **Feature Input:** Manual audio feature specification with LSTM classification
3. **Audio Upload:** Real-time audio file analysis with A/V prediction and visualization



Results

4.1 Model Performance

The BiLSTM regressor is trained using the command-line interface with configurable hyperparameters including epochs, batch size, and learning rate. The model outputs test set performance metrics including MSE and MAE for arousal-valence prediction accuracy.

Epoch-wise Metrics

Epoch	Loss	MAE	Val Loss	Val MAE	Learning Rate	Notes
1	0.0033	0.0419	7.5689e-04	0.0235	0.0010	—
2	0.0010	0.0247	4.0754e-04	0.0171	0.0010	—
3	5.6196e-04	0.0182	3.8549e-04	0.0161	0.0010	Best epoch
4	3.9918e-04	0.0154	7.5969e-04	0.0225	0.0010	—
5	3.1246e-04	0.0136	6.0041e-04	0.0201	0.0010	—
6	2.6698e-04	0.0125	5.7524e-04	0.0194	↓ LR → 0.0005	LR reduced (plateau)
7	2.0083e-04	0.0109	8.0064e-04	0.0220	0.0005	—
8	1.8378e-04	0.0104	6.5005e-04	0.0198	0.0005	—
9	1.7254e-04	0.0101	8.0564e-04	0.0217	0.0005	—
10	1.6380e-04	0.0098	7.4728e-04	0.0208	↓ LR → 0.00025	LR reduced (plateau)

The model was trained for 10 epochs using a ReduceLROnPlateau learning rate scheduler, with performance monitored on a held-out validation set. The training demonstrated rapid convergence, with the **validation loss reaching its minimum at Epoch 3 (3.85×10^{-4})** and a corresponding validation **MAE of 0.0161**. Subsequent epochs showed slight fluctuations and mild overfitting, as indicated by increasing validation loss despite continued improvements in training loss. The scheduler reduced the learning rate at **Epochs 6 and 10 to 5×10^{-4} and 2.5×10^{-4} , respectively**, in response to performance plateaus. After completion, the model weights from the best-performing epoch (Epoch 3) were restored. Final evaluation on the test dataset yielded a Test MSE of 4×10^{-4} and a Test MAE of 0.0162, confirming that the model maintained strong generalization close to its best validation performance.

4.2 Feature Representation

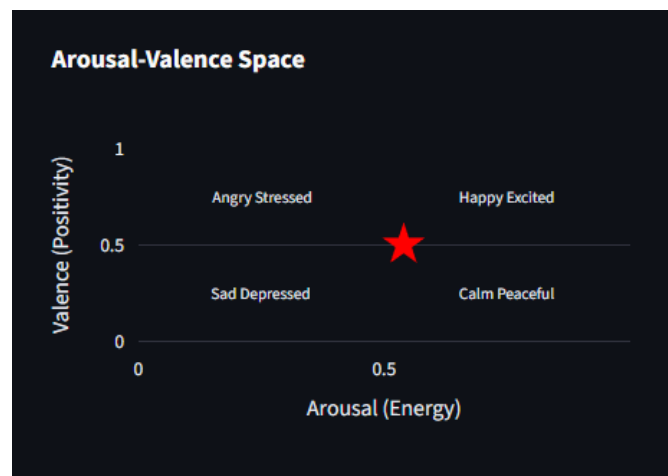
The 11-dimensional tabular feature vector successfully approximates Spotify's audio analysis features through librosa-based extraction, including energy from RMS, tempo from beat tracking, acousticness from inverse spectral flatness, danceability from onset strength, and valence from spectral centroid.

4.3 Inference Performance

The system supports both command-line inference for programmatic integration and interactive web-based inference for end-user applications. Audio files are processed in real-time, with mel-spectrogram extraction, model prediction, and visualization occurring within seconds.

4.4 User Experience

The Streamlit interface successfully visualizes emotion predictions in the arousal-valence space using Plotly plots, displays emotion quadrant classifications with descriptive text, and provides genre-specific music recommendations for therapeutic applications. The system loads multiple model types including the AV regressor, optional LSTM classifier, and baseline models for comparison.



4.5 Robustness

The feature extraction pipeline handles various audio formats (.wav, .mp3, .flac, .ogg) through soundfile library, performs automatic resampling to target sample rate, converts stereo to mono through averaging, and implements robust error handling with fallback default values.

Conclusion

This paper presented Swar_Manovigyan_ML, an emotion-aware music therapy system that bridges the gap between music emotion recognition and therapeutic applications. The system addresses key limitations of existing recommendation systems by providing continuous arousal-valence predictions rather than discrete mood categories, enabling fine-grained emotional analysis.

Our BiLSTM-based architecture successfully models temporal dependencies in audio through bidirectional processing and captures both short-term and long-term emotional patterns through multi-layer recurrent connections. The dual feature extraction pipeline supports both deep learning models (mel-spectrograms) and traditional machine learning approaches (tabular features), enabling flexible modeling strategies.

The therapeutic music recommendation system provides actionable guidance by mapping continuous A/V predictions to emotion quadrants with genre-specific suggestions tailored to emotional states. The Streamlit interface democratizes access to emotion-aware music therapy by providing an intuitive, real-time analysis tool requiring no technical expertise.

Future Work

The system's architecture supports several promising extensions:

1. **Personalization:** Fine-tuning on user-specific A/V data for individualized therapy
2. **Multi-modal Integration:** Incorporating physiological signals (heart rate, skin conductance) for multi-task learning
3. **Attention Mechanisms:** Adding attention layers over temporal frames for interpretability of emotion predictions

The open-source nature of this project enables researchers and clinicians to adapt the system for specific therapeutic contexts, contributing to the growing field of computational music therapy.

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